COM210 Engineering Maths III

Lecturer: Dr. Fotopoulos Georgios 2022/09

Quiz II

Name: _____ Student ID: _____ Marks: ____/70

Note: This quiz contains 4 questions with 70 marks in total. Answer all questions. Write your answers clearly and submit them in a single PDF file.

Question 1 (10+5+5 marks) Consider the curve

$$\mathbf{r}(t) = (t\sin t)\,\mathbf{i} + (t\cos t)\,\mathbf{j} + (t-1)\mathbf{k}, \qquad -4 \le t \le 4.$$

a. Find the unit tangent vector $\mathbf{T}$ at the point $(0, -\pi, \pi - 1)$.

b. Find $\mathbf{r}''(t)$.

c. Find $\displaystyle\int_{-\pi/2}^{\pi/2} \mathbf{r}''(t)\,dt$.

Question 2 (5+5 marks) Find the limits below, if they exist, or show that the limits do not exist.

a. $\displaystyle\lim_{(x,y)\to(1,1)} (2x^3y^4 - 4xy)$,

b. $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^4 - y^2}{x^2 + 2y^2}$.

Question 3 (10+5+5 marks) Let

$$f(x, y) = \sin(\pi xy)e^{(2x^2y-1)}.$$

a. Find the partial derivatives $f_x(x, y)$ and $f_y(x, y)$.

b. Find the directional derivative of $f(x, y)$ at the point $(1, 1)$ in the direction of the vector $\mathbf{v} = 3\mathbf{i} + 4\mathbf{j}$.

c. Find an equation of the tangent plane to the surface $z = f(x, y)$ at the point $(1, 1/2, 1)$.

Question 4 (10+10 marks)

a. Find the local maximum and minimum values and saddle points of the function

$$f(x, y) = 2x^3 + 2y^3 - 6xy.$$

b. Use the method of Lagrange multipliers to find the maximum value of the function

$$f(x, y) = 8xy + 1$$

subject to the constraint

$$x^2 + 4y^2 = 8.$$

-END OF QUIZ-